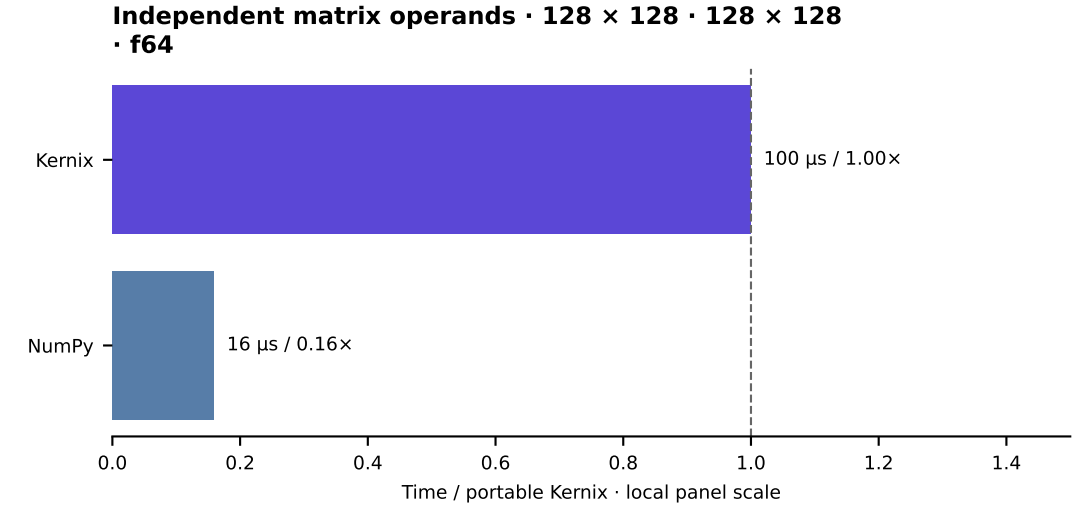
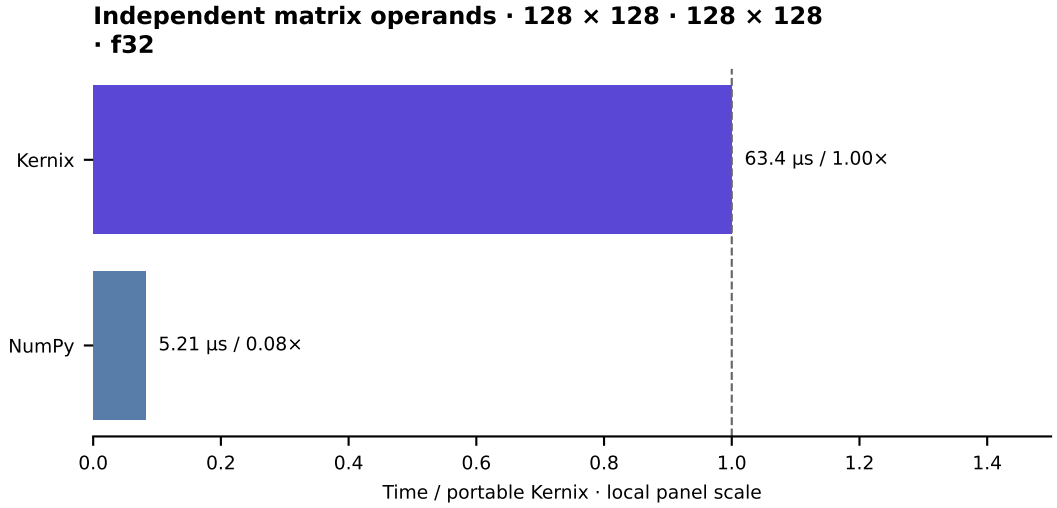
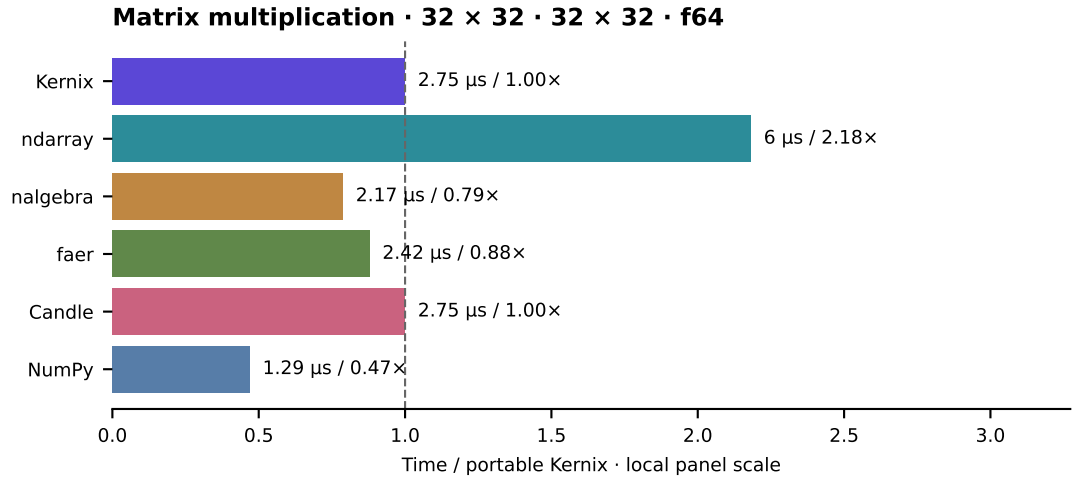
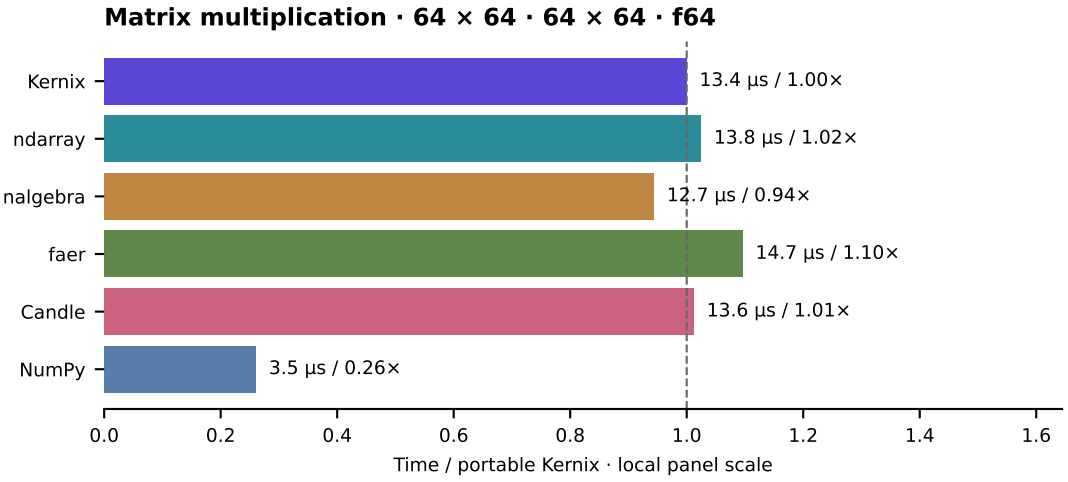
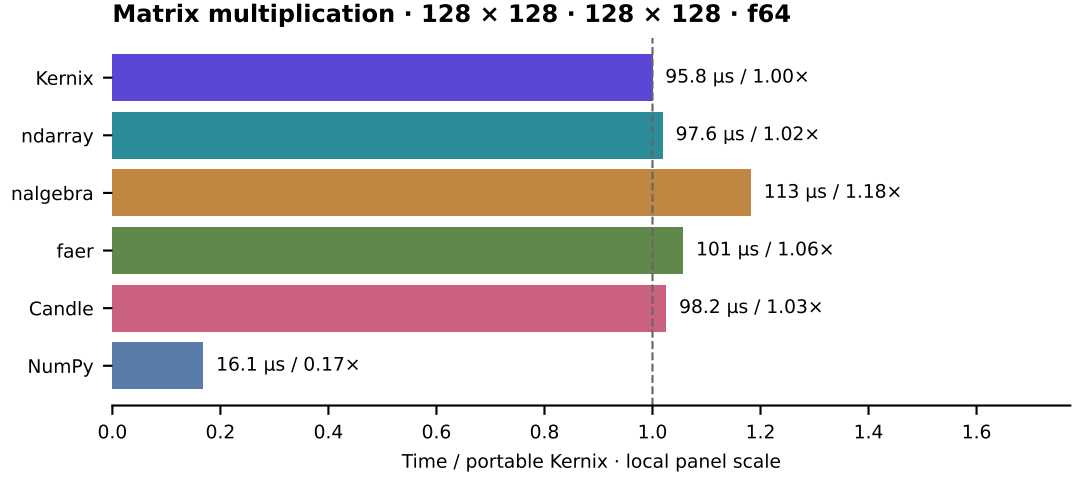
Matrix multiplication · page 1/9

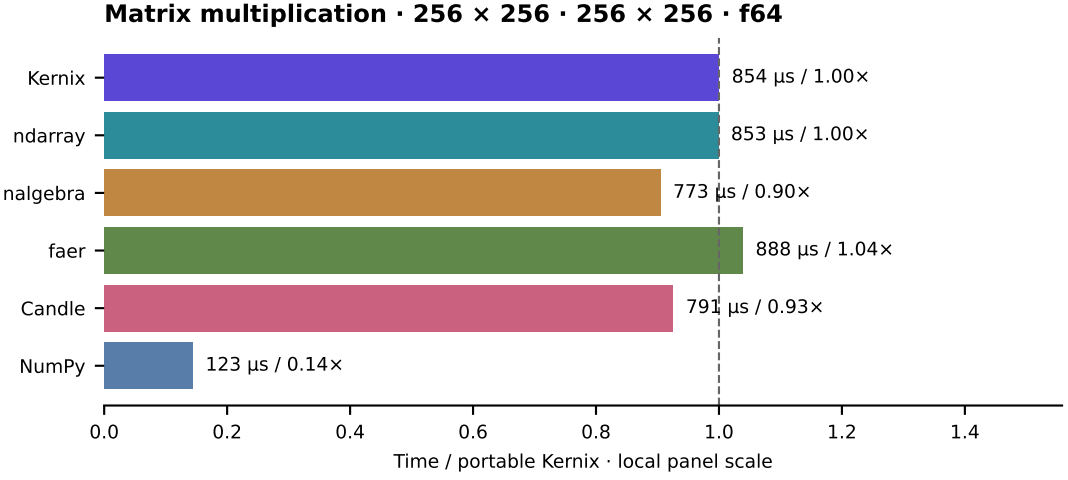




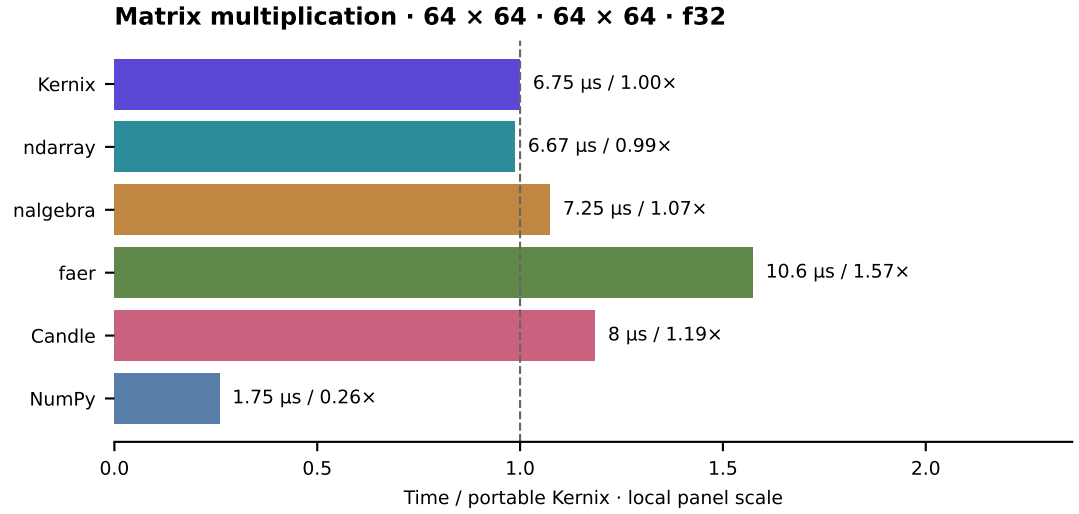


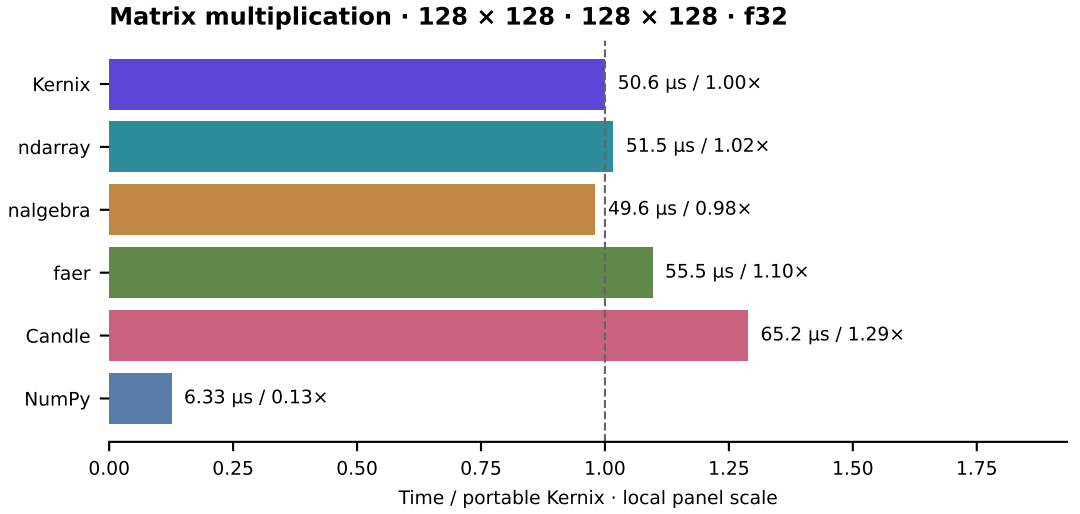


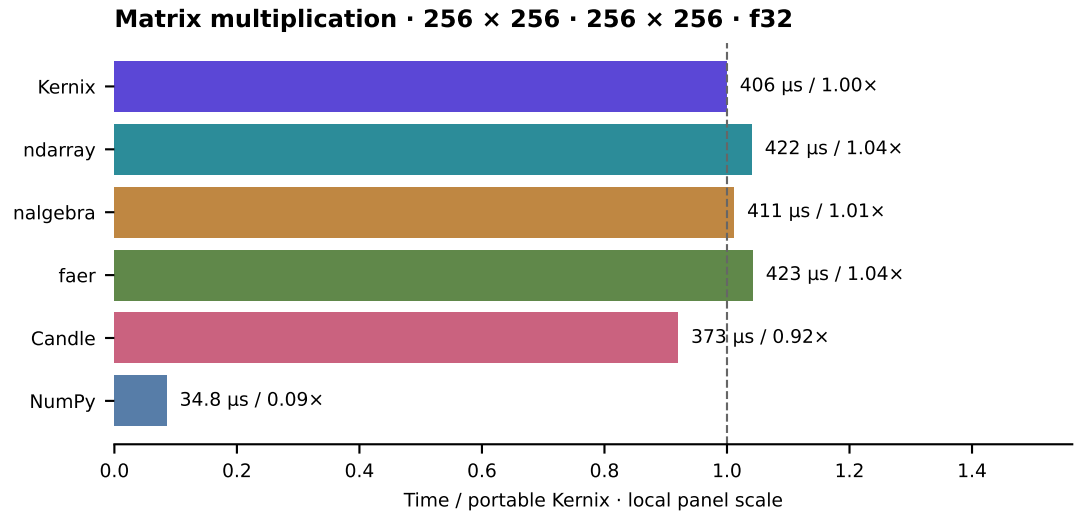


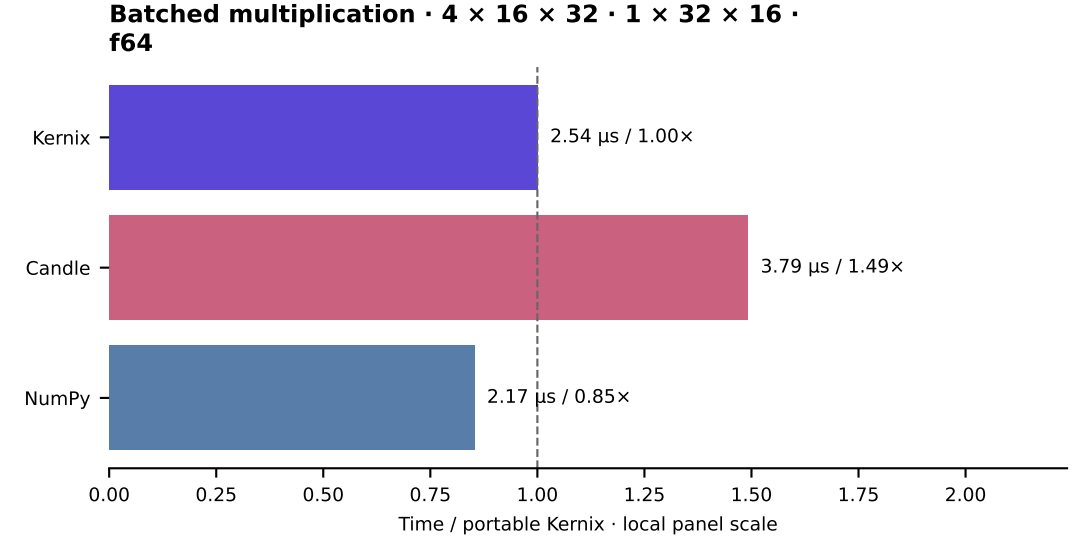


Matrix multiplication · page 2/9









Explicit parallel execution · page 3/9

Matmul · 2 workers · reused output · 64 × 64 · 64 × 64 · f64

Previous-version comparison only; no peers measured

Matmul · 2 workers · reused output · 128 × 128 · 128 × 128 · f64

Previous-version comparison only; no peers measured

Matmul · 2 workers · reused output · 256 × 256 · 256 × 256 · f64

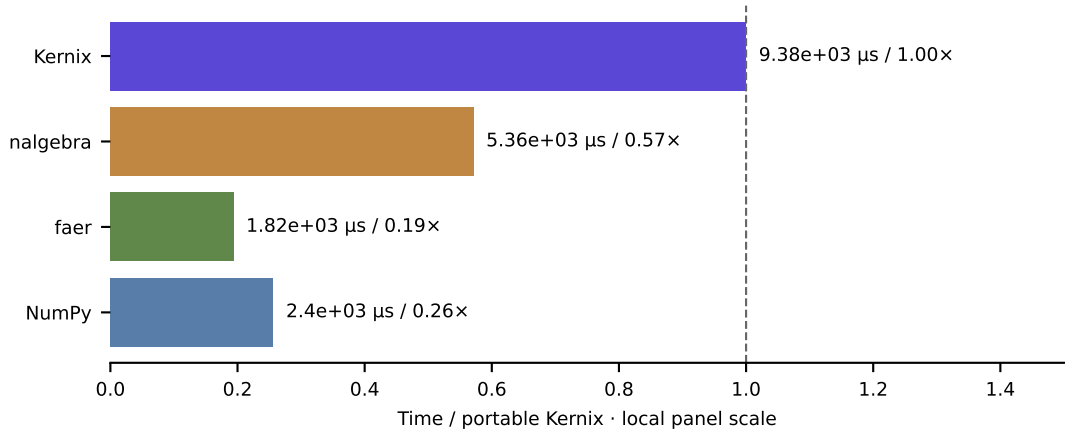
Previous-version comparison only; no peers measured

Matmul · 2 workers · reused output · 128 × 128 · 128 × 128 · f32

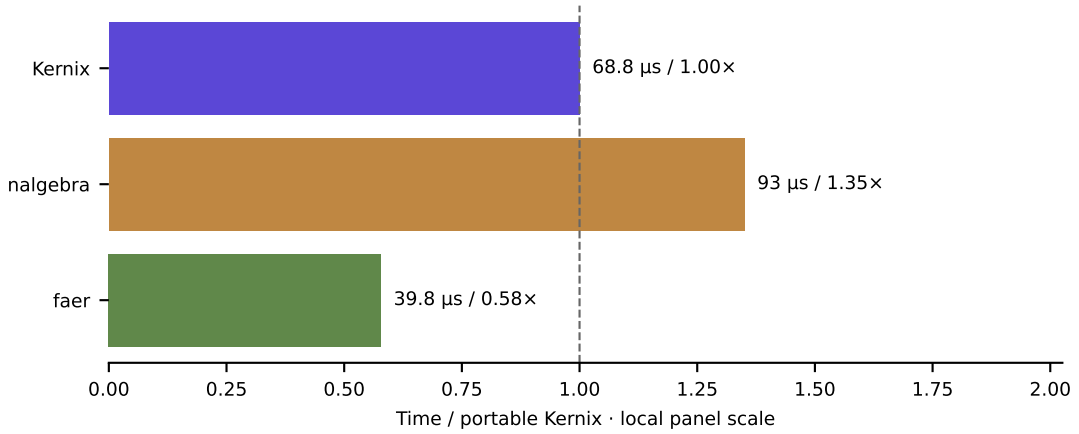
Previous-version comparison only; no peers measured

Decompositions and solvers · page 4/9

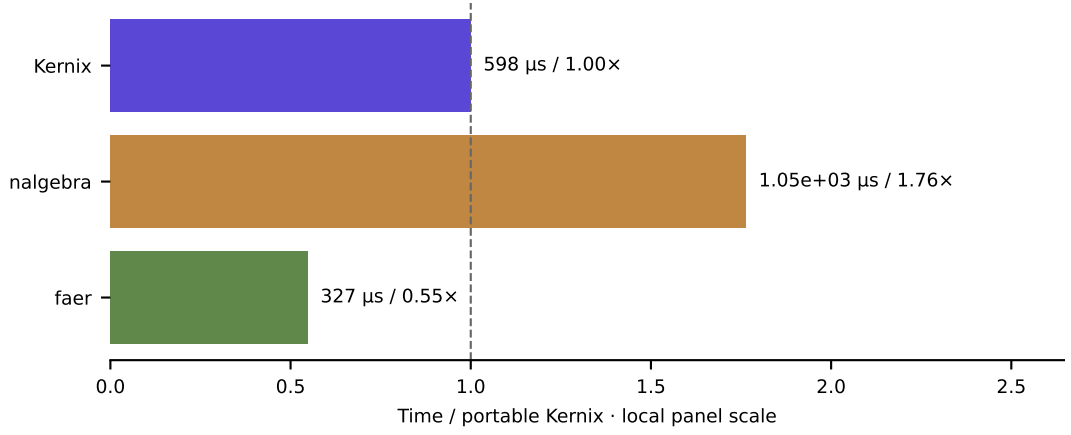
Cholesky factorization · 512 × 512 · f64



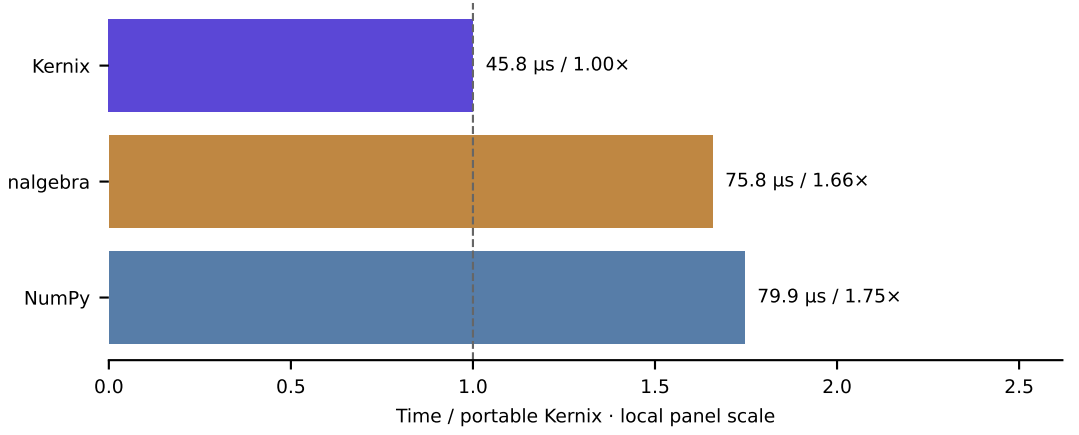
Column-pivoted QR · 96 × 32 · f64



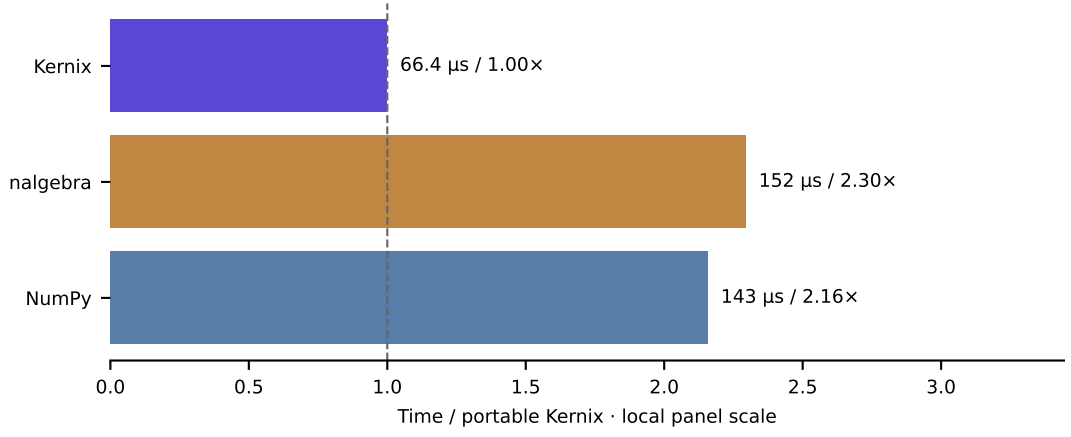
Column-pivoted QR · 256 × 64 · f64



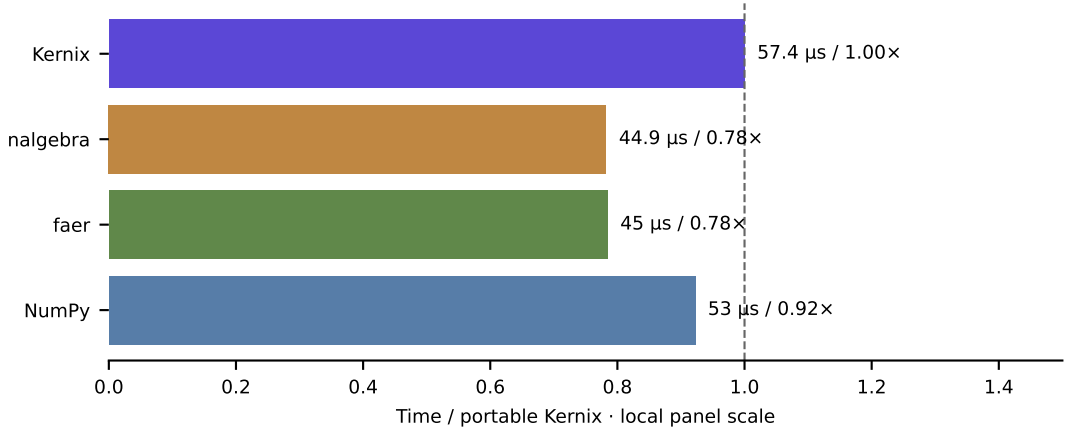
Minimum-norm least squares · 64 × 32, rank 16, 4 RHS · f64



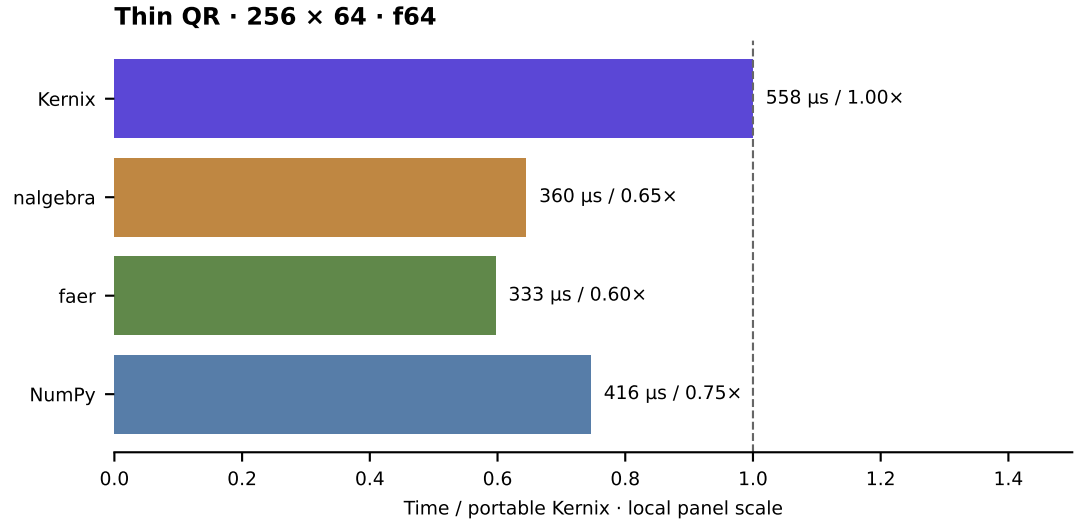
Minimum-norm least squares · 32 × 64, rank 32, 4 RHS · f64

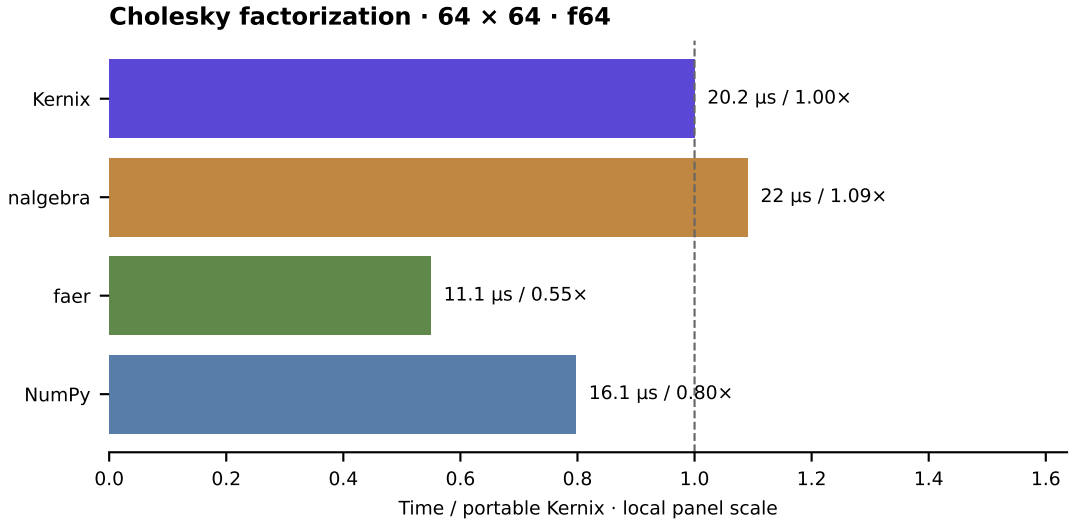


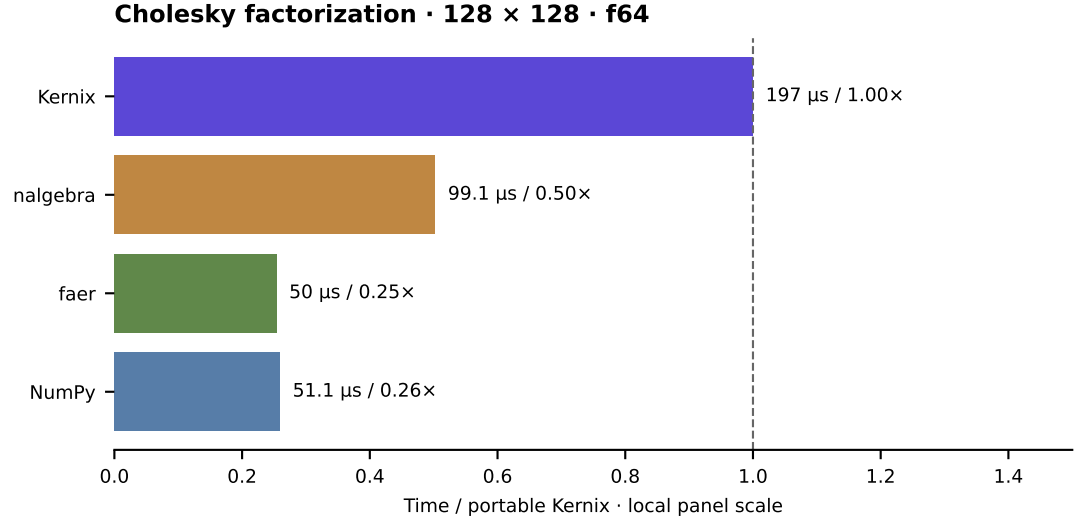
Thin QR · 96 × 32 · f64

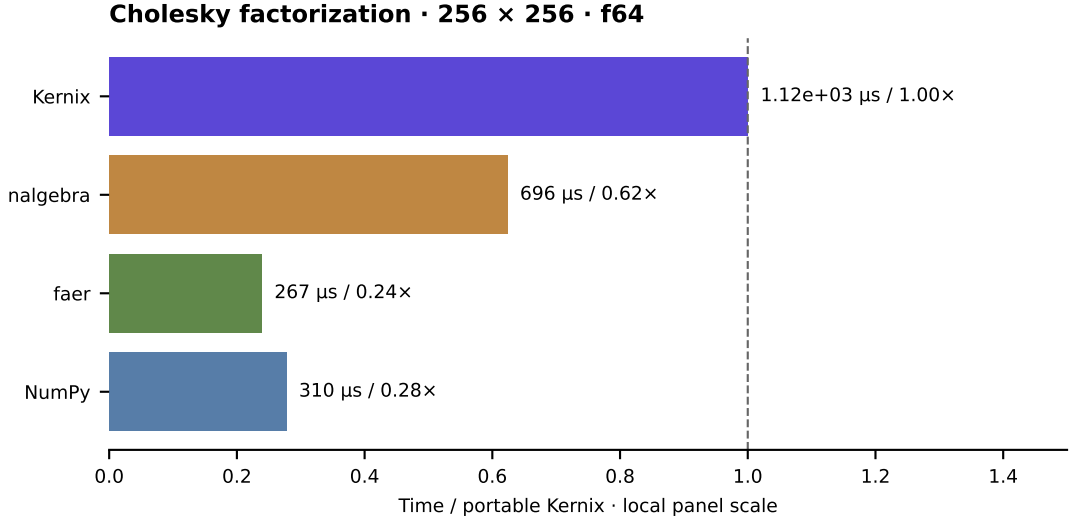


Decompositions and solvers · page 5/9

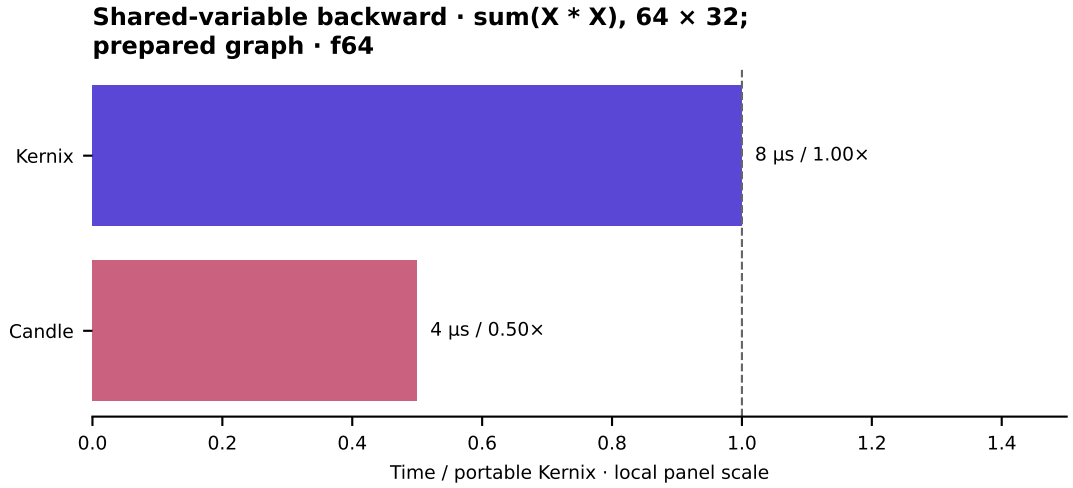


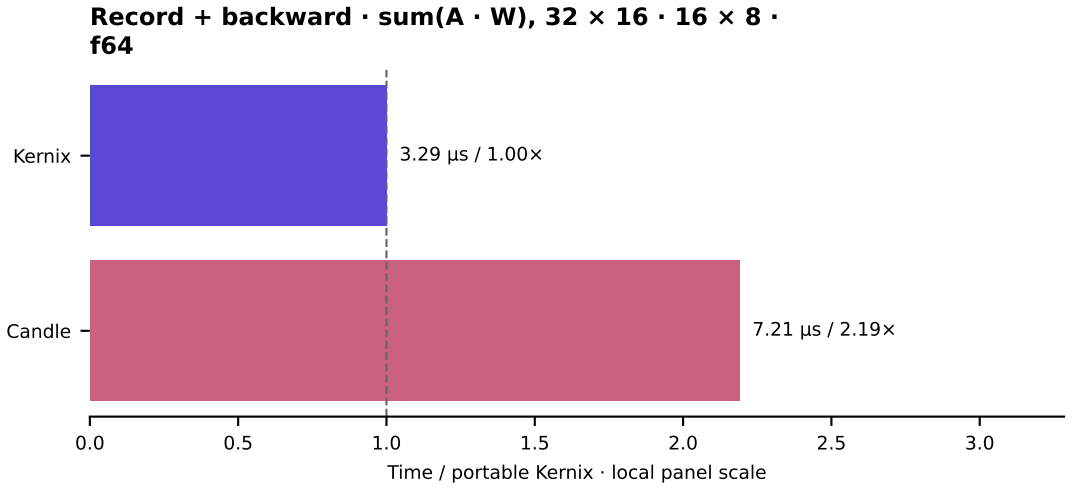


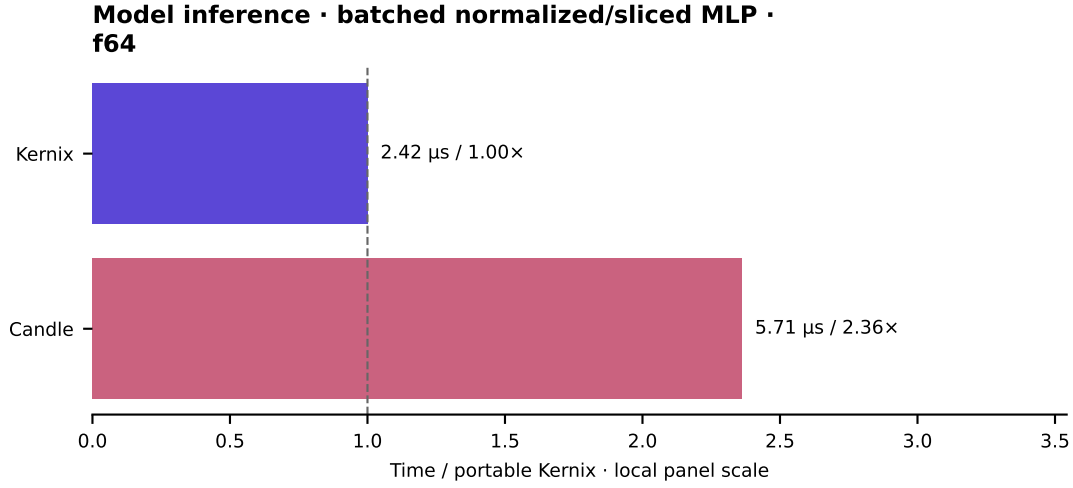


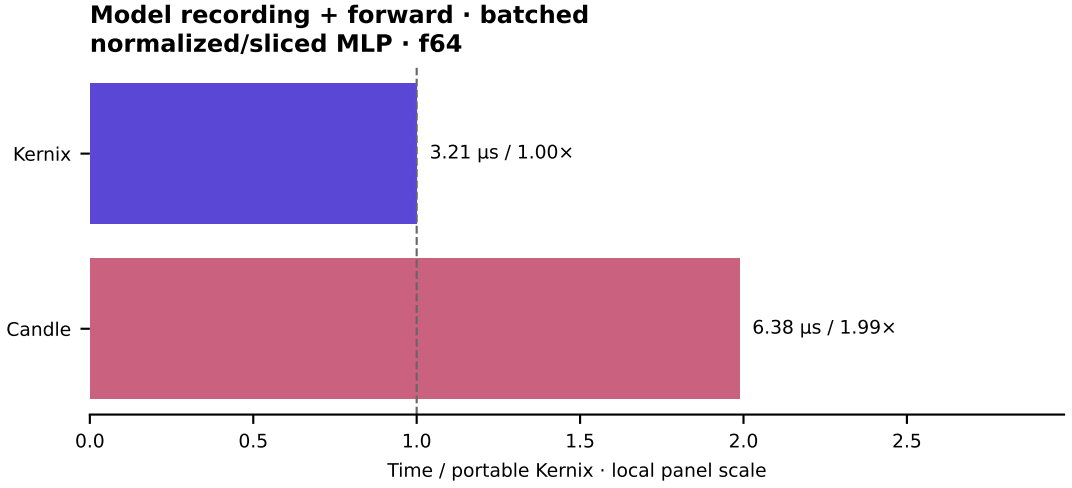


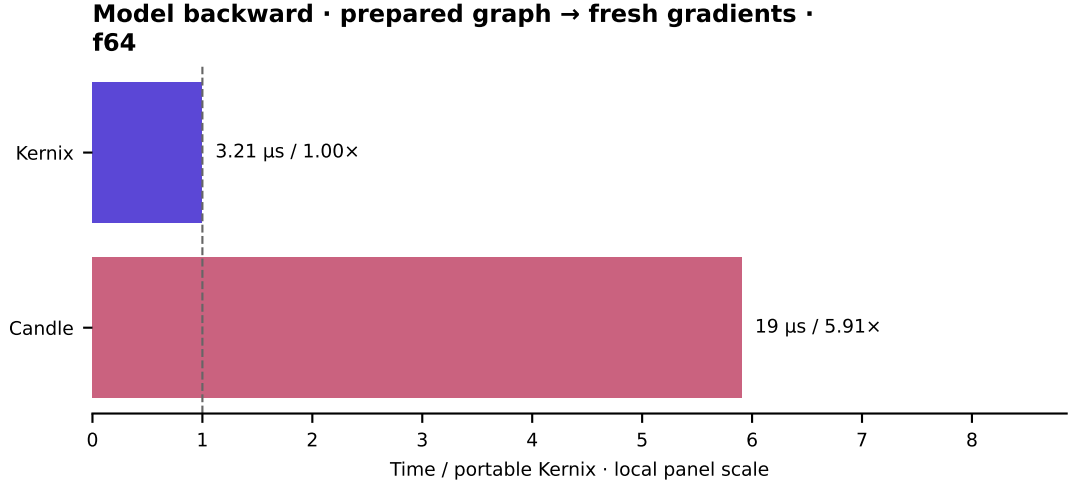
Training and autodiff · page 6/9







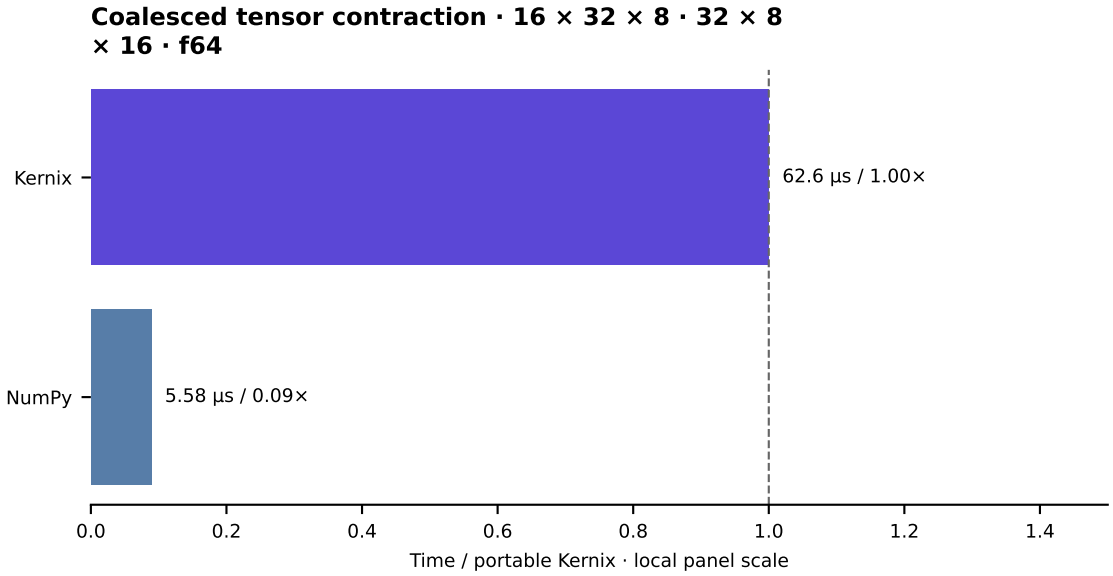


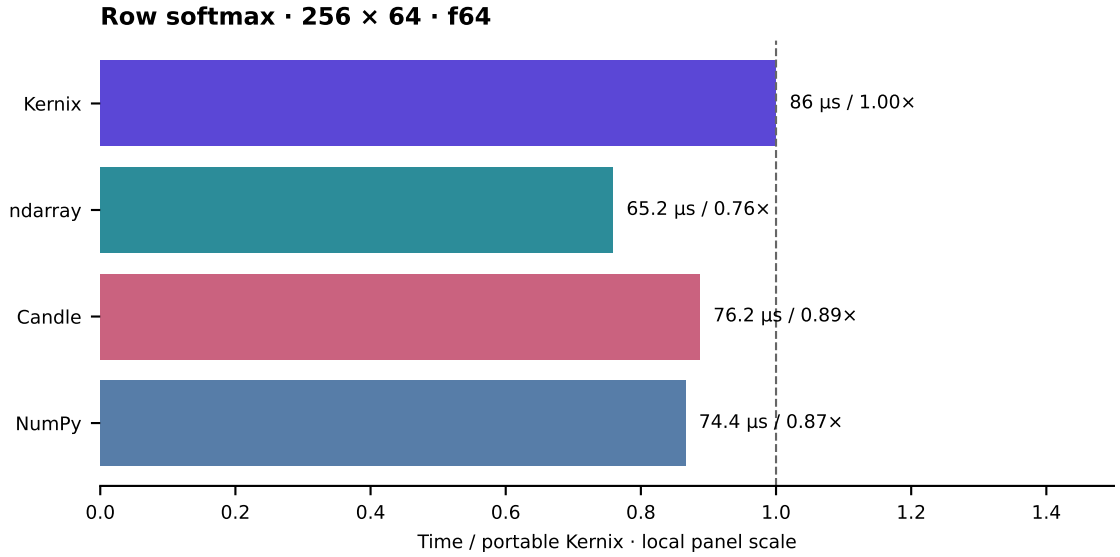


Parameter update · transactional SGD, fixed gradients · f64

Previous-version comparison only; no peers measured

Tensor contractions · page 7/9




Row softmax · 256 × 64 · f64
Kernix
ndarray
Candle
NumPy
86 µs / 1.00×
65.2 µs / 0.76×
76.2 µs / 0.89×
74.4 µs / 0.87×
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Time / portable Kernix · local panel scale

Tensor construction · page 9/9

